
EFFICIENT BAYES FACTOR SENSITIVITY ANALYSIS VIA POSTERIOR DENSITY RATIOS

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ABSTRACT

Bayes factor sensitivity analysis examines how the evidence for one hypothesis over another depends on the prior distribution. In complex models, the standard approach refits the model at each hyper-parameter value, and the total computational cost scales linearly in the grid size. We propose a method that recovers the entire sensitivity curve from a single additional model fit. The key identity decomposes the Bayes factor at any hyper-parameter value γ_x into an “anchor” Bayes factor at a fixed reference γ_0 and a Savage–Dickey density ratio in an extended model that places a hyper-prior on γ . Once this extended model is fit, the Bayes factor at any γ_x follows from the anchor value and a ratio of two posterior density ordinates. To approximate this ratio, we employ the importance-weighted marginal density estimator (IWMDE). Because the sensitivity parameter enters the model only through the prior distribution on the model parameters, the data likelihood cancels in the IWMDE, reducing it to a simple ratio of prior density evaluations on the MCMC draws, without any additional likelihood computation. The resulting estimator is fast, remains accurate even with small MCMC samples, and substantially outperforms kernel density estimation across the full sensitivity range. The method extends naturally to simultaneous sensitivity over multiple hyper-parameters and to Bayesian model averaging. We illustrate it on a univariate Bayesian t -test with exact Bayes factors for validation, a bivariate informed t -test, and a Bayesian model-averaged meta-analysis, obtaining accurate sensitivity curves at a fraction of the brute-force cost.

Keywords Savage–Dickey density ratio, prior distribution, robustness analysis, informed inference

1 Introduction

Bayes factors quantify the evidence that the data provide for one statistical model over another (e.g., Jeffreys, 1939; Kass & Raftery, 1995). For two competing hypotheses $\mathcal{H}_0$ and $\mathcal{H}_1$, the Bayes factor is defined as the ratio of the marginal likelihoods:

$$\text{BF}_{10} = \frac{p(y \mid \mathcal{H}_1)}{p(y \mid \mathcal{H}_0)}, \quad (1)$$

where $p(y \mid \mathcal{H}_1) = \int p(y \mid \xi, \mathcal{H}_1) p(\xi \mid \mathcal{H}_1) d\xi$ denotes the marginal likelihood obtained by integrating the likelihood over the prior distribution of all model parameters ξ under $\mathcal{H}_1$ (and analogously for $\mathcal{H}_0$).

The Bayes factor is therefore a ratio of prior predictives, which makes the choice of prior distribution crucial: different prior distributions encode different hypotheses and formalize different theoretical commitments about the parameters. The prior distribution should be considered an integral part of the model; if a different question is asked by specifying

a different prior distribution, a different answer ought to follow (Vanpaemel, 2010). Sources of information that can be used to construct informative prior distributions include theory, logic, previous data, and expert elicitation (Lee & Vanpaemel, 2018).

At the same time, it is important to ascertain that modest quantitative perturbations of the prior distribution do not lead to qualitatively different conclusions. One might also wish to know how much the hypothesis would need to be modified before arriving at a qualitatively different conclusion, or how the evidence depends on a specific prior choice selected from among several reasonable alternatives (e.g., Dickey, 1973; Kass & Raftery, 1995; Goodman, 1999). These questions are answered through Bayes factor *sensitivity analysis* (or robustness analysis): a range of prior distributions is specified, usually by varying the hyper-parameters of a prior distribution on one or more model parameters, and the Bayes factor is examined as a function of those hyper-parameters (e.g., Berger, 1990, 1982; Box & Tiao, 1962; Sinharay & Stern, 2002). In the robust Bayesian tradition, the general problem of prior sensitivity is formalized by specifying a class Γ of plausible prior distributions and determining the range of posterior quantities, including Bayes factors, as the prior ranges over Γ (Berger, 1990). The approach proposed here can be seen as an efficient implementation of this program for the common case where the class of prior distributions is a parametric family indexed by a hyper-parameter γ .

Bayes factor sensitivity analysis is regularly emphasized in reporting guidelines for Bayesian analyses (Andraszewicz et al., 2015; van Doorn et al., 2020; Depaoli & Van de Schoot, 2017; Kruschke, 2021) and has been historically approached from several angles. Franck & Gramacy (2020) proposed visualizing the “Bayes factor surface,” the Bayes factor plotted as a function of user-specified hyper-parameters, and used Gaussian process surrogate models to approximate these surfaces when direct computation is expensive. Johnson et al. (2023) introduced “Bayes factor functions” that express Bayes factors from common test statistics as functions of standardized effect sizes. In the context of particle physics, Fowlie (2024) proposed to plot contours of the Bayes factor over phenomenological parameters, computed through the Savage–Dickey density ratio (SDDR; Dickey, 1971; Wagenmakers et al., 2010). Support intervals (Wagenmakers et al., 2022), evidential calibration (Pawel et al., 2024), and reverse Bayes approaches (Held et al., 2022) further develop the idea of exploring the Bayes factor as a function of the prior distribution on the model parameters.

In simple models, sensitivity analyses are fast. For instance, the Bayesian t -test of Jeffreys (1948) (also see Rouder et al., 2009) involves an alternative hypothesis $\mathcal{H}_1$ under which the standardized effect size δ is assigned a Cauchy prior distribution with median 0 and scale parameter r ; consequently, the associated Bayes factor has a one-dimensional integral representation that allows rapid evaluation across a grid of r values. Alternatively, approximate Bayes factors for nested hypotheses can use Savage–Dickey normal approximation (e.g., Spiegelhalter et al., 2004; Bartoš & Wagenmakers, 2023; Dienes, 2014) which makes sensitivity analysis for a focal parameter nearly instantaneous given only a maximum likelihood estimate and its standard error (plus the prior ordinate for the test-relevant parameter at the value specified under $\mathcal{H}_0$).

For more complex models, however, Bayes factor sensitivity analysis can be prohibitively expensive. The reason is straightforward: each value of the hyper-parameter requires fitting the complete model anew. If the sensitivity analysis spans K grid points and each point requires new posterior samples, whether obtained via the SDDR, bridge sampling (Meng & Wong, 1996; Gronau et al., 2017), or nested sampling (Skilling, 2006), the total computational cost scales as K times the cost of a single fit. For models where a single fit can take from hours to days, such as model-averaged meta-analyses, structural equation models, or network models, a sensitivity analysis spanning dozens of hyper-parameter values quickly becomes infeasible (see e.g., Sekulovski et al., 2024).

In this manuscript, we combine two ideas to obtain an efficient, general-purpose method for Bayes factor sensitivity analysis that recovers the entire sensitivity curve from a single additional model fit. The first idea is to place a hyper-prior on the sensitivity parameter γ (e.g., the scale of a prior distribution) and exploit Bayes factor transitivity to decompose the target Bayes factor at any value γ_x into two components: a single “anchor” Bayes factor computed at a fixed reference value γ_0 , and a marginal likelihood ratio that is obtained via two applications of the Savage–Dickey density ratio in the hyper-parameter space. The entire sensitivity curve then follows from the posterior distribution of γ under this extended model, evaluated as a simple density ratio. The second idea is to estimate this posterior density via the importance-weighted marginal density estimator (IWMDE; Chen, 1994). Because the sensitivity parameter γ enters the model only through the prior distribution on the model parameters, the IWMDE reduces to a ratio of prior density evaluations and requires no evaluation of the data likelihood whatsoever. This property makes the estimator easy to implement whenever the prior density linking γ and the model parameters has a tractable form, and yields substantially more stable estimates than standard kernel density estimation (KDE), particularly with moderate MCMC sample sizes. We demonstrate the methodology across several examples.

2 Methods

Consider data y and two competing hypotheses $\mathcal{H}_0$ and $\mathcal{H}_1$, each specifying its own likelihood and prior distribution over its parameters. The hypotheses need not share parameters; the only requirement is that the Bayes factor $\text{BF}_{10} = p(y \mid \mathcal{H}_1)/p(y \mid \mathcal{H}_0)$ is well defined.¹

The sensitivity analysis concerns a hyper-parameter γ that governs some aspect of the prior specification under one of the hypotheses.² Write $\mathcal{H}_1(\gamma)$ for the version of the alternative hypothesis corresponding to hyper-parameter value γ , and denote the associated marginal likelihood by $Z(\gamma) = p(y \mid \mathcal{H}_1(\gamma))$. The marginal likelihood under $\mathcal{H}_0$ is $Z_0 = p(y \mid \mathcal{H}_0)$, so that $\text{BF}_{10}(\gamma) = Z(\gamma)/Z_0$. The goal is to evaluate $\text{BF}_{10}(\gamma)$ across a range $\gamma \in [\gamma_L, \gamma_U]$; the standard approach does so by computing $\text{BF}_{10}(\gamma)$ at each value independently, requiring a separate model fit at each point. In the JZS t -test of Rouder et al. (2009), for instance, the effect size δ is assigned a Cauchy prior distribution with scale r , $\mathcal{H}_1 : \delta \sim \text{Cauchy}(0, r)$, and the sensitivity analysis varies $\gamma = r$ across a range such as $[0.01, 2]$. The approach proposed here avoids this brute-force gridding through an exact probabilistic identity.

The derivation rests on two observations. First, when two point hypotheses on γ are compared, say γ_x versus γ_0 , the Bayes factor reduces to a ratio of marginal likelihoods:

$$\text{MLR}(\gamma_x, \gamma_0) = \frac{Z(\gamma_x)}{Z(\gamma_0)}, \quad (2)$$

since for point hypotheses no integration over γ is needed. Second, Bayes factors are transitive: $\text{BF}_{AC} = \text{BF}_{AB} \times \text{BF}_{BC}$ for any three hypotheses A, B, C . Combining these two observations yields

$$\text{BF}_{10}(\gamma_x) = \frac{Z(\gamma_x)}{Z_0} = \frac{Z(\gamma_x)}{Z(\gamma_0)} \times \frac{Z(\gamma_0)}{Z_0} = \text{MLR}(\gamma_x, \gamma_0) \times \text{BF}_{10}(\gamma_0), \quad (3)$$

which decomposes the target Bayes factor into two components: a marginal likelihood ratio $\text{MLR}(\gamma_x, \gamma_0)$ that captures how the evidence shifts when the hyper-parameter moves from γ_0 to γ_x , and a single “anchor” Bayes factor $\text{BF}_{10}(\gamma_0)$ that calibrates the curve against the null hypothesis (cf. Eq. 4 in Fowlie, 2024).

To compute the marginal likelihood ratio efficiently, we define an *extended model* $\mathcal{H}_\gamma$ that places a hyper-prior $\pi(\gamma)$ on the sensitivity parameter γ :

$$p(y \mid \mathcal{H}_\gamma) = \int Z(\gamma) \pi(\gamma) d\gamma. \quad (4)$$

The posterior distribution of γ under $\mathcal{H}_\gamma$ is

$$p(\gamma \mid y, \mathcal{H}_\gamma) = \frac{Z(\gamma) \pi(\gamma)}{p(y \mid \mathcal{H}_\gamma)}. \quad (5)$$

Taking the ratio of posterior densities at γ_x and γ_0 and rearranging yields the marginal likelihood ratio

$$\text{MLR}(\gamma_x, \gamma_0) = \frac{Z(\gamma_x)}{Z(\gamma_0)} = \frac{p(\gamma_x \mid y, \mathcal{H}_\gamma)}{p(\gamma_0 \mid y, \mathcal{H}_\gamma)} \times \frac{\pi(\gamma_0)}{\pi(\gamma_x)}. \quad (6)$$

Substituting into the transitivity decomposition (3) gives the main result:

$$\text{BF}_{10}(\gamma_x) = \text{BF}_{10}(\gamma_0) \times \frac{p(\gamma_x \mid y, \mathcal{H}_\gamma)}{p(\gamma_0 \mid y, \mathcal{H}_\gamma)} \times \frac{\pi(\gamma_0)}{\pi(\gamma_x)}. \quad (7)$$

The connection to the SDDR illuminates the structure of this result. For any fixed γ^* , the model $\mathcal{H}_1(\gamma^*)$ is nested within $\mathcal{H}_\gamma$, so the SDDR gives their Bayes factor as $\text{BF}_{\mathcal{H}_1(\gamma^*), \mathcal{H}_\gamma} = p(\gamma^* \mid y, \mathcal{H}_\gamma)/\pi(\gamma^*)$. Regrouping the terms in (7) accordingly reveals that the right-hand side is a product of three Bayes factors:

$$\text{BF}_{10}(\gamma_x) = \underbrace{\text{BF}_{10}(\gamma_0)}_{\text{BF}_{\mathcal{H}_1(\gamma_0), \mathcal{H}_0}} \times \underbrace{\frac{p(\gamma_x \mid y, \mathcal{H}_\gamma)}{\pi(\gamma_x)}}_{\text{BF}_{\mathcal{H}_1(\gamma_x), \mathcal{H}_\gamma}} \times \underbrace{\frac{\pi(\gamma_0)}{p(\gamma_0 \mid y, \mathcal{H}_\gamma)}}_{\text{BF}_{\mathcal{H}_\gamma, \mathcal{H}_1(\gamma_0)}}. \quad (8)$$

¹In many common testing scenarios, such as the Bayesian t -test, the two models are nested: they share a set of nuisance parameters ψ , and $\mathcal{H}_0$ is obtained from $\mathcal{H}_1$ by fixing a focal, test-relevant parameter θ to a point value θ_0 . The proposed method covers these nested cases but does not require nesting.

²For clarity, the method is presented for a scalar γ that enters $\mathcal{H}_1$; the extension to a vector γ and/or to sensitivity parameters entering $\mathcal{H}_0$ is straightforward.